In this lab sheet do **not use the C++ array class** use standard C arrays. Do not use the STL. Make sure to reference all the sources you used to solve the problem.

Classes -Safe Array

Statement: As we know in C, there is no check to determine whether the array index is out of bounds. During program execution, an out-of-bound array index can cause serious problems. Also, recall that in C++ the array index starts at 0.

Task Implementing and test (using a main program) a class called `safeArray` that solves the out-of-bound array index problem and allows the user to access the safe array contents via an .at(int index) method which takes the index position of the desired element. The .at(index) method, checks that the index is not out of bounds, if it is, it prints an error message, otherwise it returns the element at the index position.

Every object of type `safeArray` should be an array of type int. During execution, when accessing an array component, if the index is out of bounds, the program must terminate with the error message “array out of bounds” and return -1.

Make you safe array to a default size of 10 and Initialise all the elements in the `safeArray` to 0

Usage example main.cpp,

```
#include<iostream>
Using namespace std;
#include”safearray.h”

safeArray mySafeArray(); //safeArray of default size 10
safeArray anotherSafeArray(); //safeArray of default size 10
int element = 0;
cout << (element = mySafeArray(2)) << endl;
```

You must separate class declaration from definition and for this lab you will submit the following 3 files: `safearray.h safearray.cpp and main.cpp`